Rules for differentiation

1 Differentiate the following:

a $y = 3x^2$ b $y = 6x^4$ c $y = -4x^3$ d $y = 5x^5$

e $y = -\dfrac{x^3}{5}$ f $y = \dfrac{3x^6}{2}$ g $y = \dfrac{5x^4}{12}$ h $y = 8x^{0.5}$

2 Differentiate the following, expressing your answers in negative index form:

a $y = 8x^{\frac{7}{9}}$ b $y = 3x^{0.6}$ c $y = \dfrac{-2}{x^5}$ d $y = 7x^{-3}$

3 Differentiate the following, expressing your answers in surd form:

a $y = 8\sqrt{x}$ b $y = \dfrac{\sqrt{x}}{4}$ c $y = 14x^2\sqrt{x}$ d $y = \dfrac{2}{\sqrt{x}}$

4 Consider the function $y = \dfrac{7}{x}$.

a Rewrite the function in negative index form.

b Find the derivative, expressing your answer with a positive index.

5 Differentiate the following, expressing your answers in positive index form.

a $y = \dfrac{14}{x\sqrt{x}}$ b $y = \dfrac{1}{4x^3}$

6 Differentiate the following:

a $y = x - 9$ b $y = 2x + 9$

c $y = \dfrac{2x}{9} + 7$ d $y = x^2 - x + 8$

e $y = x^2 - 8x - 6$ f $y = x^4 + x^5$

g $y = x^5 - x^4 + 3$ h $y = 2x^3 - 3x^2 - 4x + 13$

i $y = \dfrac{1}{2}x^5 + \dfrac{1}{5}x^8$ j $y = -3x^5 + 5x^4 - 5x^3 - 4x^2 + 2x - 4$

k $y = \dfrac{x^8}{8} + \dfrac{x^5}{5} - 3x$ l $y = 3x^{-6} + \dfrac{x^{-3}}{7}$

m $y = \dfrac{24}{x^5} - \dfrac{30}{x^4}$ n $y = x^{\frac{1}{2}} + 8x^{\frac{3}{4}}$

7 Differentiate $y = 7ax^7 - 2bx^3$, where a and b are constants.

8 Differentiate the following. Write your ans[illegible] with positive indices.

a $y = x^3 + x^{-5} - 9$

b $y = \sqrt[3]{x} + \dfrac{1}{x^7} - 4$

c $y = \dfrac{1}{\sqrt[4]{x}} - x^7 + \pi$

d $y = x^3\sqrt{x} + 3x^5$

9 Consider the function $f(r) = \dfrac{2}{r} + \dfrac{r}{3}$.

a Rewrite the function so that each term is a power of r.

b Find $f'(r)$.

10 Differentiate $y = \dfrac{2}{x^a} - \dfrac{3}{x^b}$, where a and b are constants.

11 For each of the following functions:

i Rewrite the function in expanded form.

ii Differentiate the function.

a $y = \dfrac{4}{9}(-4x - 8)$

b $y = (6x + 5)(x + 3)$

c $y = 2x^2(7x + 2)$

d $y = x(3x + 4)(5x + 6)$

e $y = (x + 4)^2$

f $y = 5(x - 3)^2$

g $y = (8x - 4)^2$

h $y = \left(2x + \dfrac{3}{x}\right)(6\sqrt{x} + 5)$

12 Differentiate the following functions:

a $y = (3x + 2)(7x + 6)$

b $y = (x + 8)(x - 7) + 5$

13 Consider the function $f(x) = \left(\sqrt{x} + 10x^2\right)^2$

a Rewrite the function $f(x)$ in expanded form, with all terms written as powers of x.

b Hence, differentiate the function.

14 Consider the following functions:

i Rewrite the function so that each term is a power of x.

ii Differentiate the function.

a $f(x) = \dfrac{8x + 3}{2x}$

b $y = \dfrac{8x^9 - 4x^8 + 6x^7 + 9}{2x^2}$

c $y = \dfrac{3x^6 + 9x^4 + \sqrt{x} + 11}{x^3}$

d $y = \dfrac{8x^2 + 6x + 4}{\sqrt{x}}$

15 Consider the function $y = \dfrac{5x\sqrt{x}}{4x^5}$.

a Rewrite the function in simplified negative index form.

b Find $\dfrac{dy}{dx}$.

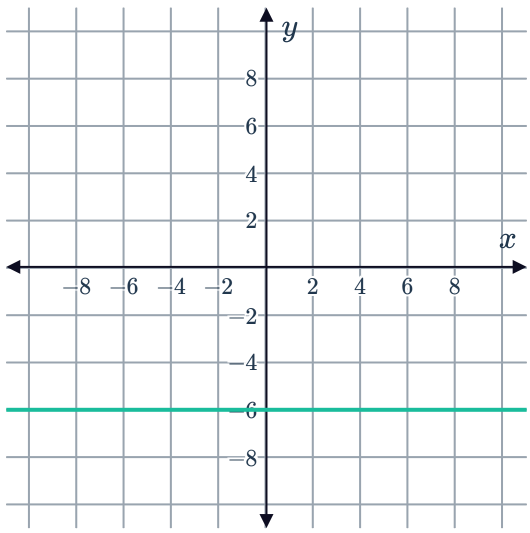
16 Differentiate the following functions, expressing your answers with positive indices.

a $y = \dfrac{x^9 + 1}{x^5}$ b $y = \dfrac{x^2 + x^{13}}{x^7}$ c $y = \dfrac{x + 1}{\sqrt[7]{x}}$ d $y = \dfrac{\sqrt{x^3} + 5x}{\sqrt[3]{x}}$

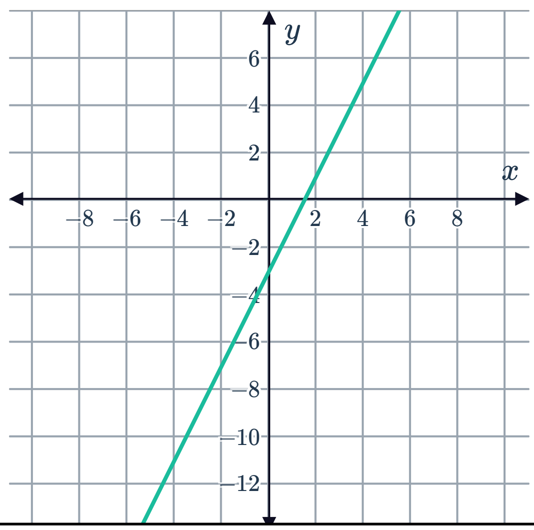
Gradients

17 Find $f'(2)$ if $f'(x) = 4x^3 - 3x^2 + 4x - 6$.

18 Consider the graph of $f(x) = -6$ shown:
Find $f'(4)$.

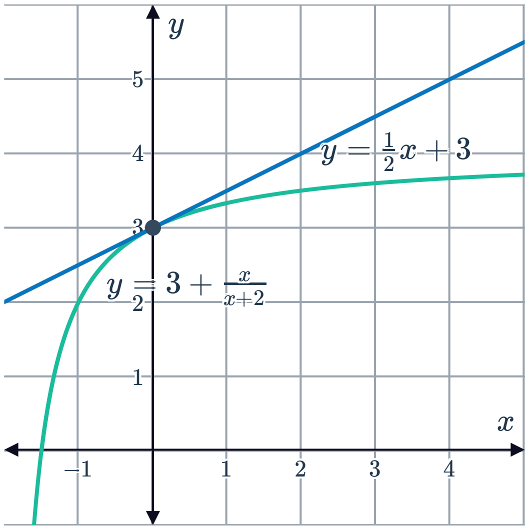


19 Consider the graph of $f(x) = 2x - 3$:
Find $f'(-4)$.



20 The tangent to the curve $y = 3 + \dfrac{x}{x+2}$ at the point $(0, 3)$ has the equation $y = \dfrac{1}{2}x + 3$:

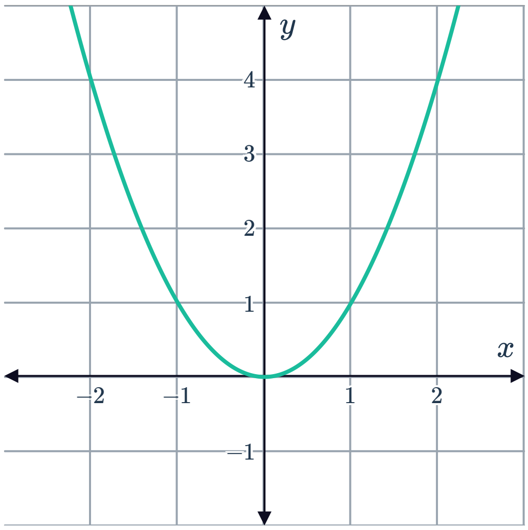
Find $f'(0)$.



21 Consider the graph of the function $f(x) = x^2$:

a How many points on the graph of $f(x) = x^2$ have a gradient of 2?

b Find the x-coordinate of the point at which $f(x) = x^2$ has a gradient of 2.



22 Find the gradient of the following functions at the given point:

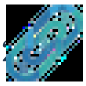
a $f(x) = 16x^{-3}$ at the point $(2, 0)$.

b $f(x) = x^4 + 7x$ at the point $(2, 30)$.

c $f(x) = \dfrac{6}{\sqrt{x}}$ at the point $\left(25, \dfrac{6}{5}\right)$.

d $f(x) = x^3 - 2x^4 + \sqrt{x}$ at the point $(4, -446)$.

e $f(x) = \dfrac{11}{x} + \dfrac{10}{x^2}$ at the point $\left(4, \dfrac{27}{8}\right)$.

23 Consider the function $f(x) = 6x^2 + 5x +$ 

a Find $f'(x)$.

b Find $f'(2)$.

c Find the x-coordinate of the point at which $f'(x) = 41$.

24 Consider the function $f(x) = x^3 - 4x$.

a Find $f'(x)$.

b Find $f'(4)$.

c Find $f'(-4)$.

d Find the x-coordinates of the points at which $f'(x) = 71$.

25 Consider the function $y = 2x^2 - 8x + 5$.

a Find $\dfrac{dy}{dx}$.

b Hence, find the value of x at which the gradient is 0.

26 Find the x-coordinates of the points at which $f(x) = -3x^3$ has a gradient of -81.

27 Find the x-coordinate of the point at which $f(x) = \sqrt{x}$ has a gradient of 6.